# Infrared Remote Signal Collection



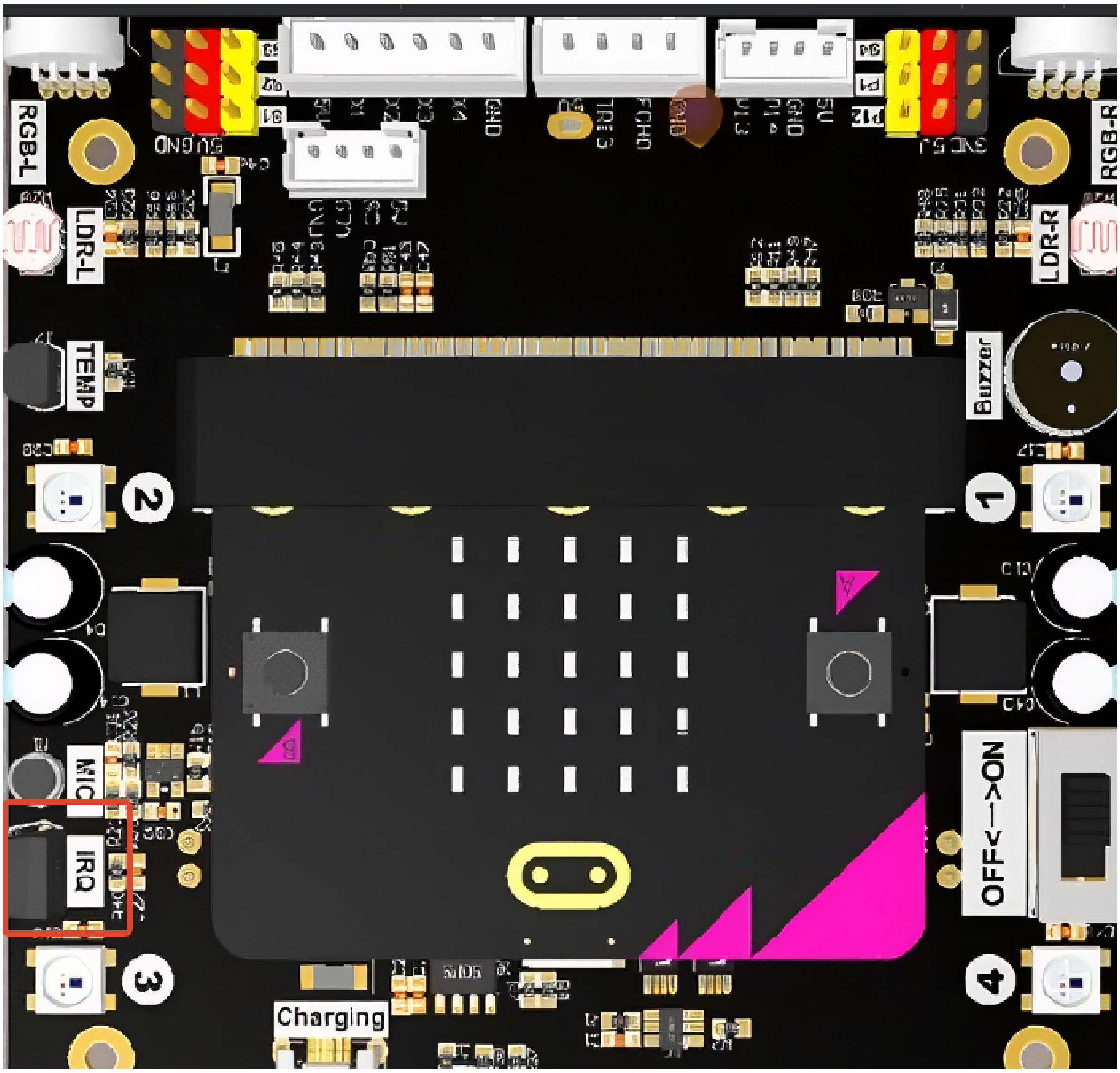
## Experimental Purpose

The Lastbit car is equipped with an HS0038B infrared receiving module, marked with the silkscreen IRQ. The HS0038B is a highly integrated **38kHz infrared remote control receiving module**. When a key on the infrared remote control is pressed, it drives the infrared LED to flash at high speed (invisible to the human eye). After the infrared receiving probe identifies the infrared signal, the internal circuit of the module outputs a level signal after optical filtering, demodulation, and shaping. It can be used with an infrared remote control (NEC protocol). In this section, we will learn how to read the infrared remote control keys and perform display or control functions based on different keys.

With the standard NEC protocol, each time a key is pressed, **4 bytes (32 bits)** of data are sent, with the following structure:

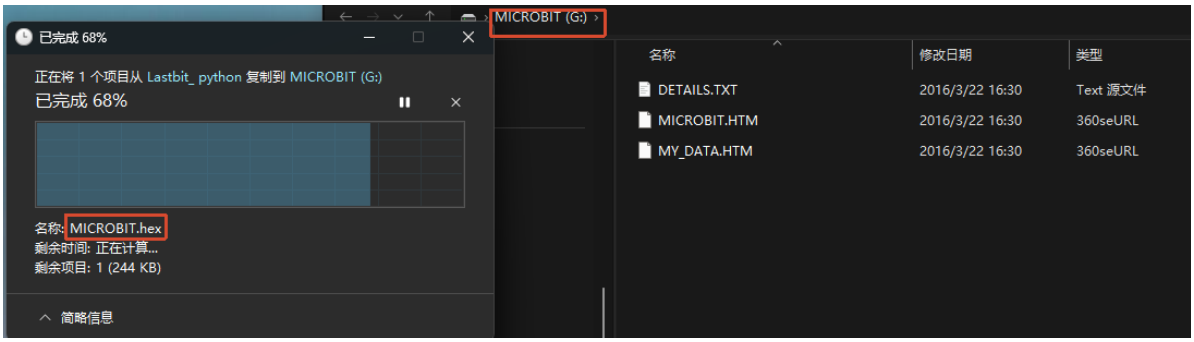
```
Guide code → Address code → Address inverse code → Command code → Command inverse code → End bit
```

- **Address code**: identifies the device type (such as TV, DVD) to prevent different devices from interfering with each other.
- **Address inverse code (~Address)**: the address code inverted bit by bit, used to verify whether the address transmission is correct.
- **Command code**: the specific key command (such as volume +, power off).
- **Command inverse code (~Command)**: the command code inverted bit by bit, used to verify whether the command transmission is correct.




## Experimental Steps

**Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.**




1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**

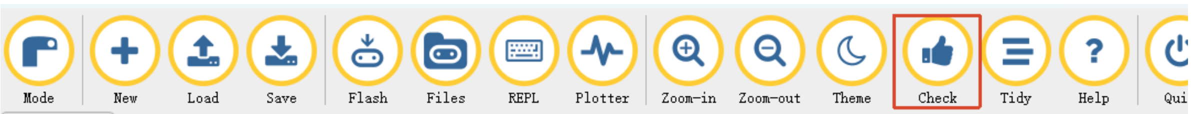
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

   Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.


3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

**Whether or not you save does not affect the code check and flashing.**

4. At this point, you can click `Check` to check whether the code format has any problems.




5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.


6. Unplug the Microusb data cable and switch the switch to the ON position.

## Code Implementation

```
# 7.Infrared_Remote_Signal_Collection

from microbit import *
from lastbit import LASTBIT
import music

# Import the infrared remote control module
import ir

Lastbit = LASTBIT()
```

```
NUM_CUR = 0

# Initialize the infrared remote control module
ir.init(pin8)

display.show(Image.YES)

while True:

    # Read the infrared remote control signal
    value = ir.read(pin8)
    value = value >> 8

    # When the infrared remote control signal value is not -1, it indicates that
a key
    if value != -0x01:
        if value == 0x80:    # ↑
            display.show(Image.ARROW_S)
        elif value == 0x90:  # ↓
            display.show(Image.ARROW_N)
        elif value == 0x20:  # ←
            display.show(Image.ARROW_E)
        elif value == 0x60:  # →
            display.show(Image.ARROW_W)
        elif value == 0x10:  # ↖
            display.show(Image.ARROW_SE)
        elif value == 0x50:  # ↗
            display.show(Image.ARROW_SW)

        elif value == 0x30:  # +
            NUM_CUR += 1
            display.show(NUM_CUR)
        elif value == 0x70:  # -
            NUM_CUR -= 1
            display.show(NUM_CUR)

        elif value == 0x00:  # POWER
            display.scroll("Power")

        elif value == 0xA0:  # BEEP
            display.scroll("Beep")

        elif value == 0x40:  # LIGHT
            display.scroll("Light")

        elif value == 0x08:  # 1
            NUM_CUR = 1
            display.show(1)
        elif value == 0x88:  # 2
            NUM_CUR = 2
            display.show(2)
        elif value == 0x48:  # 3
            NUM_CUR = 3
            display.show(3)
        elif value == 0x28:  # 4
            NUM_CUR = 4
            display.show(4)
        elif value == 0xA8:  # 5
```

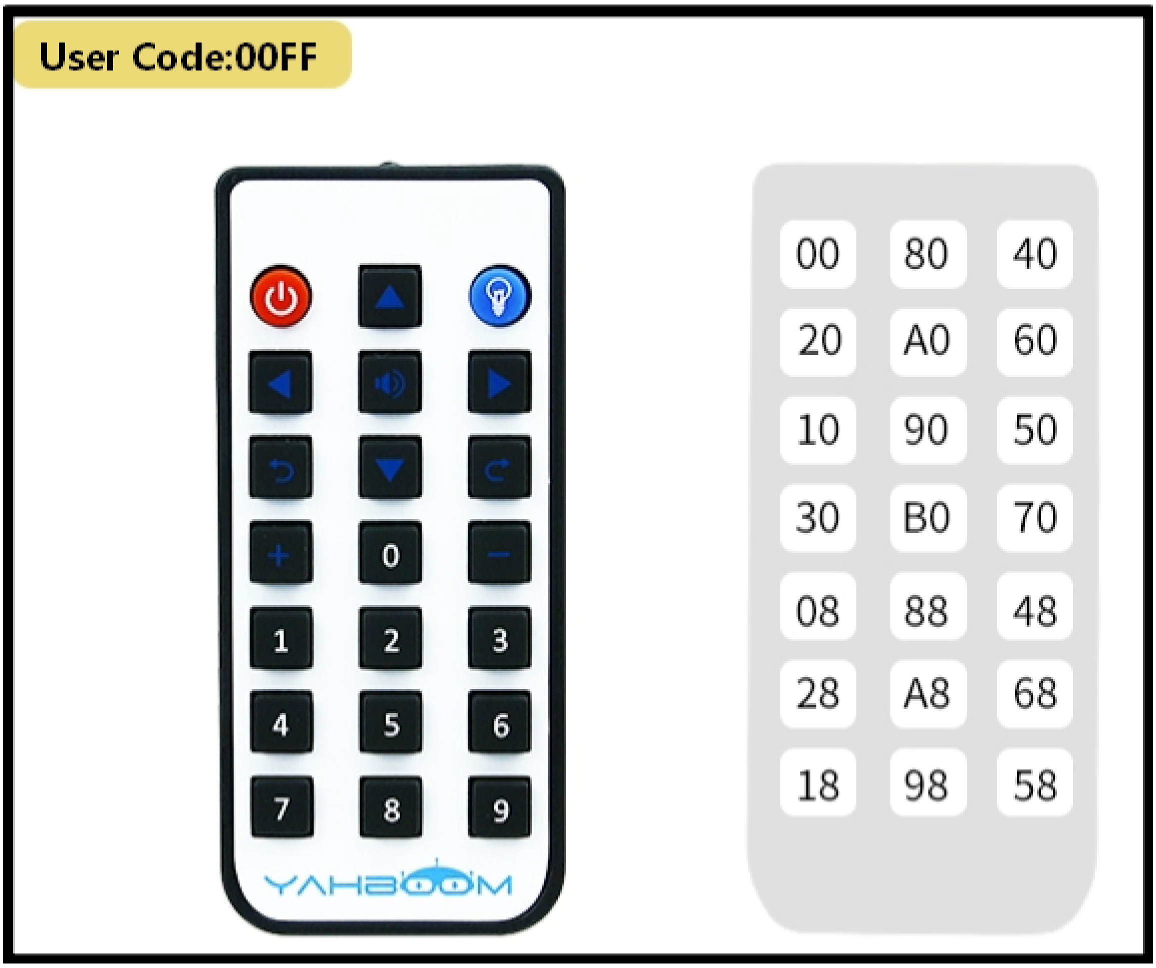
```
            NUM_CUR = 5
            display.show(5)
        elif value == 0x68:  # 6
            NUM_CUR = 6
            display.show(6)
        elif value == 0x18:  # 7
            NUM_CUR = 7
            display.show(7)
        elif value == 0x98:  # 8
            NUM_CUR = 8
            display.show(8)
        elif value == 0x58:  # 9
            NUM_CUR = 9
            display.show(9)
        elif value == 0xB0:  # 0
            NUM_CUR = 0
            display.show(0)

    sleep(10)
```

`ir.read(pin8)`: reads the key value of the button data returned by the infrared remote control.

# Experimental Phenomenon

After flashing the program, turn on the switch (switch it to the ON position), and a check mark will be displayed on the Microbit LED matrix. At this time, we can use the remote control to control the car.

**The remote control has a plastic film on the bottom when it leaves the factory. It needs to be torn off before use. It is best to point it directly at the infrared receiving probe for control. When using the remote control, it is recommended to press the keys briefly and wait for a while between each press before pressing again.**



Functions corresponding to the infrared remote control keys

| Remote Control (Key Value) | Actual Function in This Tutorial's Code |
|---|---|
| 0x00 | Displays "Power" |
| 0x80 | Displays "↑" |
| 0x40 | Displays "Light" |
| 0x20 | Displays "←" |
| 0xA0 | Displays "Beep" |
| 0x60 | Displays "→" |
| 0x10 | Displays "↖" |
| 0x90 | Displays "↓" |
| 0x50 | Displays "↗" |
| 0x30 | Controls the car to speed up; the number on the screen increases by 1 each time it is pressed |
| 0x70 | Controls the car to slow down; the number on the screen decreases by 1 each time it is pressed |
| 0xB0 | Displays the number 0 |
| 0x08 | Displays the number 1 |
| 0x88 | Displays the number 2 |
| 0x48 | Displays the number 3 |
| 0x28 | Displays the number 4 |
| 0xA8 | Displays the number 5 |
| 0x68 | Displays the number 6 |
| 0x18 | Displays the number 7 |
| 0x98 | Displays the number 8 |
| 0x58 | Displays the number 9 |
| 0xFF | Handle user code (no function, only used to identify the remote control) |